

Hybrid Rare-Event Estimation for Conjunction Risk: Benchmark Results and Fault-Tolerant Resource Sketches

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Repository: <https://github.com/Xyloth/xpqa-rare-event-qae>

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Abstract

Satellite conjunction assessment relies on repeated probability-of-collision (P_c) estimation under uncertainty as new tracking updates arrive. For most conjunctions, coarse screening is sufficient, but decisions near maneuver thresholds require rapid time-to-confidence in tail probability, where brute-force Monte Carlo can become expensive and sensitive to distributional assumptions. This report presents a reproducible encounter-plane benchmark suite that stress-tests rare-event estimators across baseline and near-threshold regimes, including mixture (non-Gaussian) uncertainty modes. We compare naive Monte Carlo, importance sampling, and a splitting-style rare-event estimator using confidence-interval width versus sample budget as the primary metric. We then outline how the same event-predicate structure enables quantum amplitude-estimation-style acceleration in a hybrid workflow, and we provide initial fixed-point precision and fault-tolerant resource sketches to bound feasibility assumptions. Code and scripts to reproduce all figures are available in the accompanying repository.

Keywords

conjunction assessment; collision probability; rare-event estimation; importance sampling; splitting; amplitude estimation; fault-tolerant quantum computing

1. Introduction

Spacecraft operators routinely assess conjunction risk by estimating the probability of collision (P_c) under uncertainty and updating that estimate as new tracking data arrive. In practice, most conjunctions can be cleared with coarse screening, but a small fraction fall into a decision-sensitive “gray zone” where small changes in state, covariance, or modeling assumptions can materially alter the risk estimate and the recommended action. In these cases, operators may request additional tracking, perform higher-fidelity analysis, and—if thresholds are exceeded—coordinate and execute collision avoidance maneuvers. The operational goal is therefore not simply to compute a probability once, but to reach a decision with defensible confidence under time pressure.

This report focuses on the computational bottleneck that appears in precisely that gray zone: estimating rare-event probabilities with tight confidence intervals using sampling-based methods. Brute-force Monte Carlo is straightforward and widely applicable, but its accuracy improves slowly as sample count grows, which becomes costly when P_c is small or when the decision hinges on narrowing uncertainty near a threshold. In addition, real-world uncertainty can be imperfectly captured by a single Gaussian model; mixture (non-Gaussian) modes—such as a low-probability, higher-variance component—can dominate tail risk while remaining largely invisible to coarse screening. These effects motivate benchmark-driven evaluation of rare-event estimators in regimes that are operationally relevant but statistically challenging.

The contribution of this technical note is twofold. First, we provide a reproducible encounter-plane benchmark suite for conjunction-style probability estimation that includes (i) baseline scenarios and (ii) near-threshold stress tests, with both single-mode and mixture uncertainty models. We compare three classical estimators—naive Monte Carlo, importance sampling, and a splitting-style rare-event estimator—using confidence-interval (CI) width versus sample budget as the primary metric, with additional analysis of a simple gated refinement policy that reflects how operational pipelines avoid over-spending compute on clearly safe cases. Second, we outline how the same event-predicate structure supports a hybrid classical–quantum workflow in which quantum amplitude-estimation-style routines are invoked only in the refinement slice, and we provide initial feasibility sketches in terms of fixed-point precision requirements and fault-tolerant resource estimates.

We emphasize scope. The intent is not to replace mature operational systems or to claim near-term deployment on noisy quantum hardware. Rather, the goal is to (a) make the rare-event estimation bottleneck concrete and measurable on well-defined instance families; (b) identify regimes where estimator choice strongly affects time-to-confidence; and (c) establish a transparent pathway for evaluating when fault-tolerant quantum resources could offer an

advantage for refinement workloads under explicit oracle and state-preparation assumptions. The accompanying repository contains code and scripts to reproduce all figures.

2. Problem framing and decision boundary

This work models conjunction risk in the common “encounter plane” abstraction: at the time of closest approach, uncertainty in the relative position of two objects is represented as a distribution in a local 2D plane perpendicular to relative motion. A collision (or hard-body contact) occurs when the relative position falls within a circular hard-body radius around the origin of that plane. Under these assumptions, the collision probability $\mathbf{Pc}$ is the probability mass of the relative-position distribution inside the hard-body region.

Operationally, conjunction pipelines rarely require the same level of effort for every event. Most conjunctions are obviously safe under coarse screening. The expensive cases are those near action thresholds, where confidence and calibration matter: small changes in state estimate, covariance, or modeling assumptions can move an estimate across a decision boundary and trigger additional tracking, coordination, or maneuver planning. We therefore evaluate estimators not just by point accuracy, but by the rate at which they tighten confidence intervals—i.e., time-to-confidence under a fixed evaluation budget.

Benchmark tiers (event rarity regimes)

We define two rarity regimes to stress estimator behavior in the tail:

- **Tier 2 (moderate tail):** “rare but observable.” Naive Monte Carlo produces nonzero collision-event counts at typical sample budgets, enabling straightforward estimation and CI construction.
- **Tier 3 (extreme tail):** “ultra-rare.” Under typical budgets, naive Monte Carlo frequently yields **zero hits**, making estimates unstable and confidence intervals uninformative without variance reduction.

The purpose of these tiers is not to claim a universal operational frequency, but to probe the practical boundary where estimator choice becomes decisive.

Uncertainty models: single-mode vs. mixture

In addition to a single-mode uncertainty model (a single distribution describing the relative-position uncertainty), we include **mixture** uncertainty cases. The motivation is simple: even if a wide-error mode has small weight, it can dominate tail probability and materially change $\mathbf{Pc}$ while remaining difficult to detect under coarse screening. This “rare mode dominates tail risk” behavior is a plausible source of decision surprises in real pipelines and is a natural stress test for rare-event estimators.

3. Benchmark design

Event predicate and evaluation budget

Each estimator ultimately relies on repeated evaluation of the same event predicate:

Event: the sampled encounter-plane relative position falls inside the hard-body region.

In our experiments, the cost metric is the number of event-predicate evaluations (samples) required to achieve a target confidence interval width. This isolates the statistical efficiency of each estimator independent of implementation details, and it aligns with the way both classical and quantum sampling-based methods scale (number of oracle or event checks).

Scenario families: baseline and near-threshold

We evaluate two scenario families:

1. **Baseline scenarios:** representative configurations within Tier 2 and Tier 3 regimes, intended to show estimator scaling behavior as rarity increases.
2. **Near-threshold scenarios:** “decision boundary” stress tests constructed so that the mean miss distance lies near the hard-body threshold, making the tail mass highly sensitive to small shifts. This regime is designed to emulate the operational gray zone where additional computation is commonly justified.

Metrics and reporting

We report uncertainty primarily via confidence-interval width as a function of sample budget N . For each estimator and scenario, we compute CI statistics across replicated runs (or an equivalent resampling procedure) to capture variability in rare-event regimes.

4. Classical baselines

We compare three sampling-based estimators:

4.1 Naive Monte Carlo (MC)

Naive Monte Carlo draws samples directly from the target uncertainty model and estimates $\mathbf{Pc}$ as the fraction of samples that satisfy the event predicate. MC is unbiased and simple, but it becomes inefficient as events become rare: achieving a fixed absolute error typically requires sample counts that grow rapidly, and in Tier 3 it often produces zero-hit runs at realistic budgets.

4.2 Importance sampling (IS)

Importance sampling modifies the sampling distribution to draw more frequently from the dangerous region and then reweights samples to maintain an unbiased probability estimate. When a good proposal distribution is available, IS can reduce variance by orders of magnitude relative to naive MC. However, IS performance depends strongly on proposal quality; poor proposals can produce unstable weights and negate the advantage.

4.3 Splitting-style rare-event estimator (SPLIT)

The splitting-style estimator increases the effective number of rare-event hits by progressively filtering samples through intermediate thresholds, allocating more sampling effort toward trajectories (or samples) that approach the rare-event region. This family includes multilevel splitting and related subset-style strategies. The goal is to convert an ultra-rare event into a sequence of less-rare conditional events that can be estimated more efficiently.

4.4 Gated refinement policy (compute allocation)

Finally, we evaluate a simple gated policy that reflects real operational structure: most cases are screened cheaply, and only the subset that falls within a configurable margin of the threshold is

sent to refinement. We report expected event-evaluation counts as a function of the screening margin, compared against an “always refine” proxy policy.

This policy analysis is not presented as an operational directive; it is included to show how estimator improvements translate into end-to-end workload reductions when embedded in a realistic coarse-to-fine workflow.

5. Results

This section summarizes estimator performance on the encounter-plane benchmark suite, emphasizing confidence-interval (CI) width as a function of sample budget and focusing on two operationally motivated stressors: (i) ultra-rare tail regimes where naive Monte Carlo becomes zero-hit at typical budgets, and (ii) decision-boundary scenarios where small model changes can swing conclusions.

5.1 Baseline scenarios (Tier 2 and Tier 3)

Across baseline encounter-plane scenarios, variance-reduction methods consistently tighten confidence intervals faster than naive Monte Carlo at the same sample budget. In Tier 2 (“rare but observable”), naive MC remains viable, but both importance sampling and the splitting-style estimator reduce CI width substantially for a given N , indicating faster time-to-confidence. In Tier 3 (“ultra-rare”), the contrast becomes decisive: naive MC frequently yields zero-hit runs at typical budgets, causing unstable estimates and uninformative confidence intervals, whereas importance sampling and especially splitting-style estimation remain informative and continue to tighten uncertainty with increasing N . This is the regime where estimator choice determines whether a probability estimate is usable at all.

Mixture uncertainty cases amplify this effect. When a low-probability wide-error mode is present, tail probability can be dominated by mixture components that are poorly explored under naive sampling. In our suite, mixture scenarios generally increase estimator sensitivity to proposal mismatch, which appears as degraded performance for poorly tuned importance sampling while splitting-style approaches remain comparatively robust. The takeaway is not that one estimator universally dominates, but that distributional realism (including non-Gaussian modes) is a practical driver of tail risk and therefore a meaningful stress test.

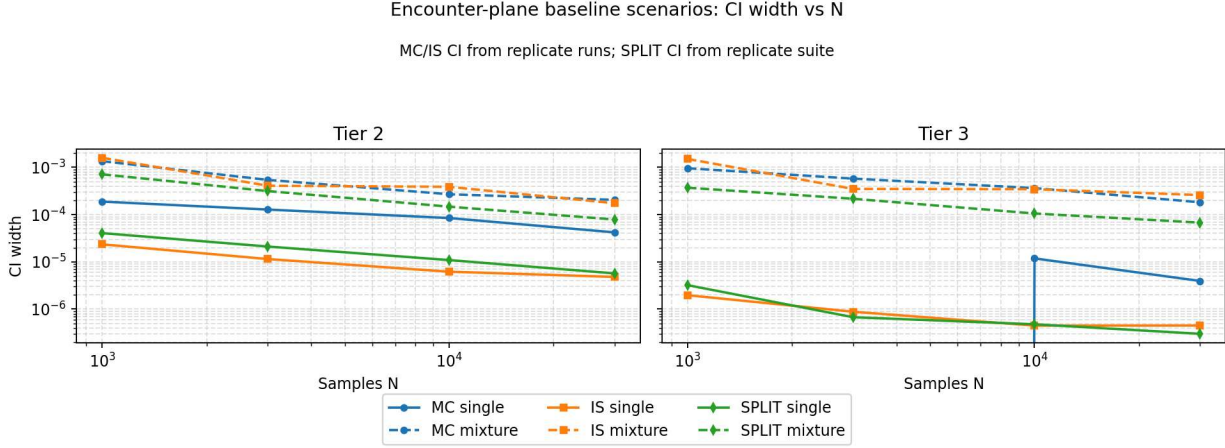


Figure 1. Confidence-interval width versus sample budget for baseline scenarios in Tier 2 and Tier 3 regimes. Curves compare naive Monte Carlo (MC), importance sampling (IS), and the splitting-style estimator (SPLIT), for both single-mode and mixture uncertainty models. Tier 3 highlights the regime where naive MC becomes unstable due to zero-hit runs, while IS and SPLIT remain informative.

5.2 Near-threshold (decision-boundary) stress tests

Near-threshold scenarios are constructed to emulate the operational gray zone: the mean miss distance lies near the hard-body boundary, so small changes in state, covariance, or distributional assumptions can materially affect $\mathbf{P}_c$ and therefore action recommendations. In this regime, estimator performance is best interpreted as time-to-confidence rather than point-estimate accuracy alone.

In our results, near-threshold cases expose sharper differences between estimators and between single-mode versus mixture assumptions. When the proposal distribution aligns well with the rare-event region, importance sampling can deliver rapid CI tightening. When proposal mismatch occurs—particularly under mixture models or near-boundary geometry—importance sampling can become erratic, reflecting weight instability. Splitting-style estimation generally remains more stable because it adapts effort via intermediate thresholds rather than relying on a single global proposal. These observations motivate a workflow in which refinement effort is reserved for gray-zone conjunctions and, within that slice, method choice (and potentially hybrid quantum insertion) is informed by diagnostics of model mismatch.

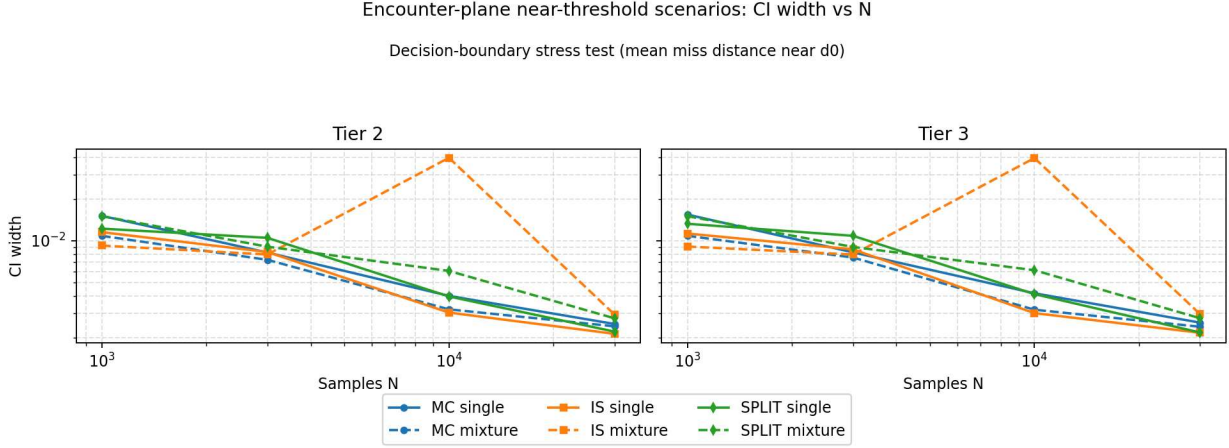


Figure 2. Confidence-interval width versus sample budget for decision-boundary stress tests (mean miss distance near the hard-body threshold). This regime captures threshold sensitivity where small modeling changes can flip conclusions, emphasizing time-to-confidence rather than point estimates alone. Mixture cases illustrate how non-Gaussian modes can stress proposal-based estimators.

5.3 Compute allocation via coarse-to-fine gating

Operational systems avoid over-investing compute on obviously safe cases by screening cheaply and refining only when an event falls near a risk boundary. To translate estimator gains into end-to-end workload impact, we evaluate a simple gated refinement policy that triggers refinement when screening outputs fall within a configurable margin around the decision boundary. The expected number of event evaluations per scenario falls substantially under moderate gating margins compared with an always-refine proxy, and the relationship is tunable: increasing the margin increases the refined fraction (and compute), while decreasing the margin reduces compute but risks deferring refinement for borderline cases.

This analysis supports the hybrid architecture assumed throughout this report: classical screening dominates overall throughput, and refinement—where rare-event estimation methods (and later, quantum routines) can be applied—is concentrated on the small subset of conjunctions that are both operationally relevant and statistically hard. In our scenario-population simulation ($K=2000$), the screen classified approximately 53.1% SAFE, 36.8% DANGER, and 10.1% GRAY; refining only GRAY reduced expected event-evaluation compute by ~88% versus an always-refine proxy.

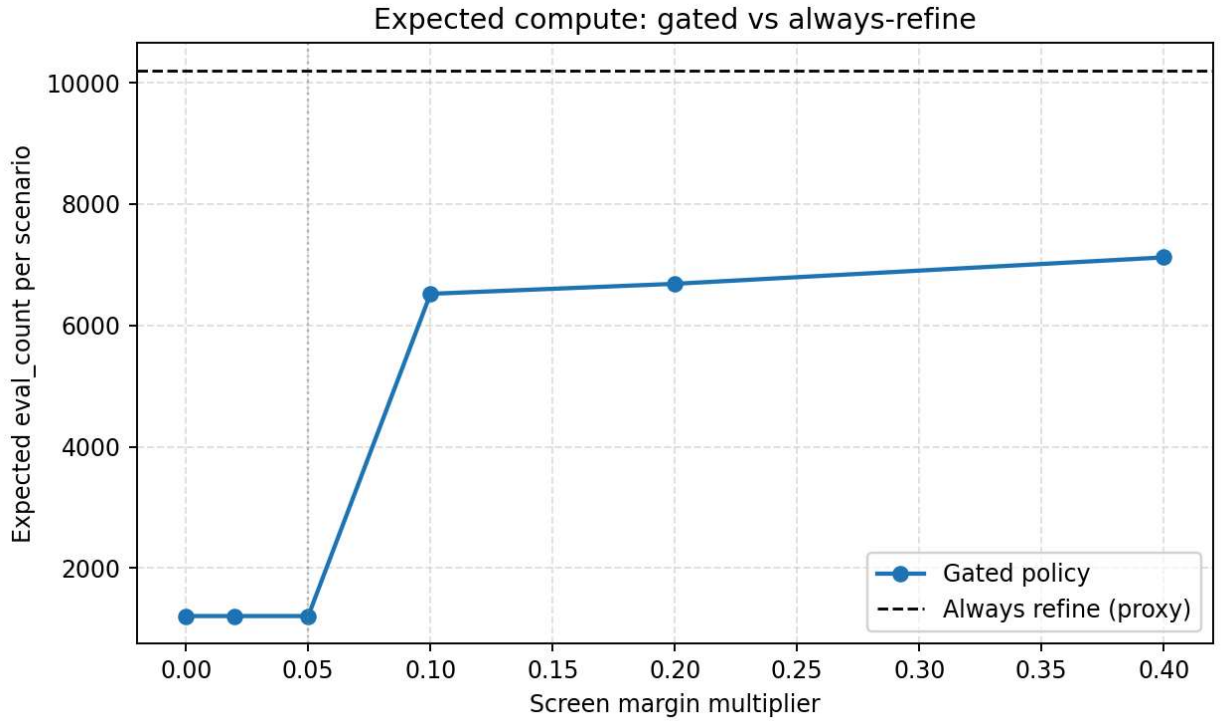


Figure 3. Expected event-evaluation count per scenario for a coarse-to-fine gated policy as a function of the screening margin multiplier, compared to an always-refine proxy. The plot illustrates how compute scales when refinement is reserved for threshold-sensitive cases, supporting a hybrid screening→refinement workflow.

6. Quantum mapping (oracle + state preparation)

This report’s quantum contribution is deliberately narrow and operationally grounded: we do not propose a fully quantum conjunction pipeline. Instead, we map a well-defined refinement subtask—rare-event probability estimation under an encounter-plane uncertainty model—onto a standard quantum sampling primitive. The intent is to identify when (and under what assumptions) fault-tolerant quantum resources could reduce the number of expensive event-predicate evaluations needed to reach decision-grade confidence.

6.1 Collision probability as an expectation value

In the encounter-plane model, collision probability can be written as an expectation under the uncertainty distribution of the relative position (x, y) at time of closest approach. Let d_0 denote the hard-body radius (or chosen safety threshold). Then:

$$\mathbf{Pc} = \mathbf{E}[\mathbf{I}(\mathbf{x}^2 + \mathbf{y}^2 < \mathbf{d0}^2)]$$

where $\mathbf{I}(\cdot)$ is an indicator: it equals 1 when the condition is true (“hit”) and 0 otherwise (“no hit”). This framing is convenient because it makes the classical and quantum tasks directly comparable: both approaches repeatedly evaluate an event predicate under a distribution and estimate its expectation.

6.2 Oracle definition (event predicate)

In the quantum mapping, the core subroutine is an oracle that implements the event predicate reversibly. Conceptually, the oracle takes a representation of a sampled encounter-plane state (x , y) (and any auxiliary parameters) and flips a designated flag bit if the predicate is satisfied:

- **Input:** (x , y) drawn from the encounter-plane uncertainty model
- **Output:** a single “hit/no-hit” flag indicating whether $\mathbf{x}^2 + \mathbf{y}^2 < \mathbf{d0}^2$

The practical lesson is that oracle cost dominates feasibility. A conjunction workflow that requires full orbit propagation inside the oracle would be an immediate non-starter. The mapping we use here is intentionally “physics-lite” and reversible-friendly: fixed-point arithmetic, bounded precision, and a simple threshold predicate. This keeps the oracle aligned with the benchmark suite and makes resource accounting tractable.

6.3 State preparation (loading uncertainty)

The second required component is state preparation: a procedure that loads samples from, or a coherent superposition over, the encounter-plane uncertainty distribution. In the simplest case, this corresponds to preparing ($\mathbf{x}, \mathbf{y}$) according to a discretized approximation of the target distribution. In more realistic settings, state preparation must reflect mixture uncertainty modes and correlation structure, which can substantially increase cost. For this reason, we treat state preparation explicitly as part of the end-to-end resource story rather than assuming it is free.

6.4 Where amplitude estimation fits

Classically, Monte Carlo estimates $\mathbf{Pc}$ by repeated independent sampling and predicate evaluation, and its error decreases slowly with sample budget. Quantum amplitude-estimation-style routines target the same expectation value but can achieve a more favorable dependence on the target additive error $\mathbf{eps}$ in terms of the number of coherent oracle invocations, under standard access assumptions for state preparation and the oracle.

The purpose of this section is not to promise near-term speedups on noisy hardware. Rather, it is to establish the testable hypothesis that, once fault-tolerant quantum resources are available and oracle/state-prep costs are bounded, amplitude estimation can reduce the number of expensive predicate evaluations needed to achieve a specified precision—making it a plausible refinement coprocessor in gray-zone conjunction analysis.

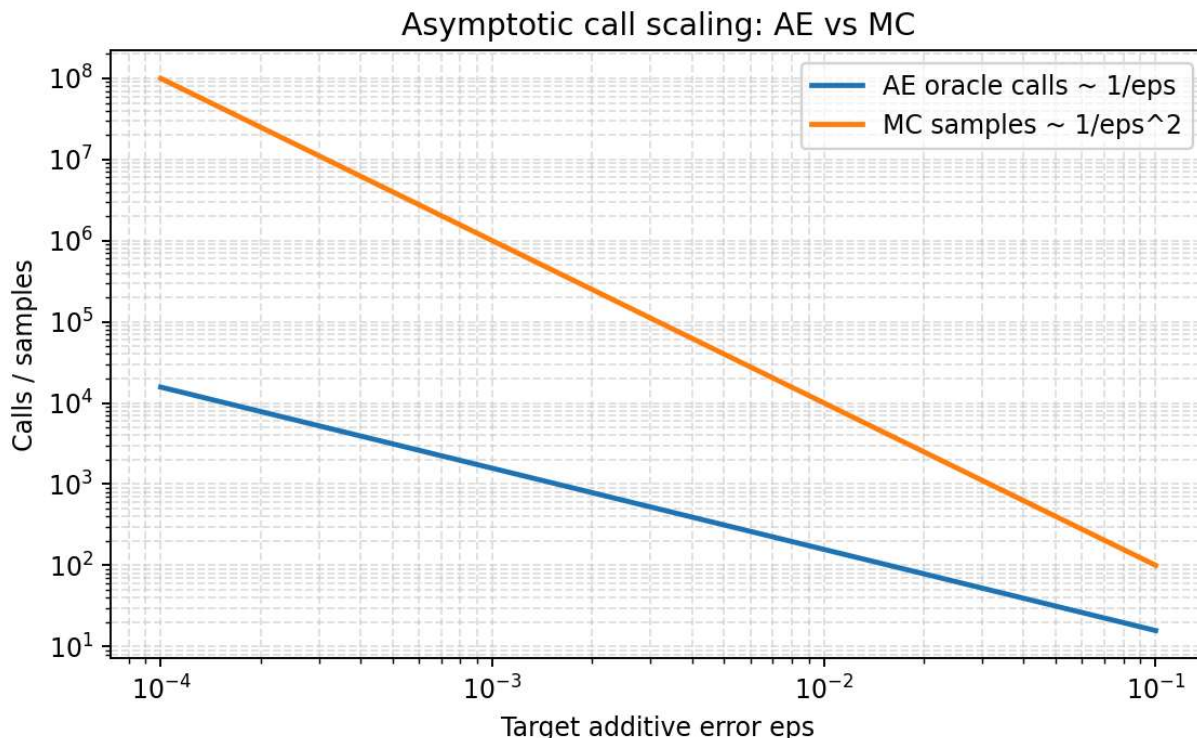


Figure 4. Asymptotic sampling/query scaling for amplitude-estimation-style routines versus naive Monte Carlo as a function of target additive error **eps**. This motivates why quantum sampling primitives are relevant specifically in high-precision or tail-regime estimation, assuming a feasible oracle and state-preparation implementation.

7. Feasibility: precision and resource sketches

The hybrid framing in this report separates two questions. The first is statistical: how many event-predicate evaluations are required to reach decision-grade confidence in tail regimes? Sections 3–5 address that empirically. The second is feasibility: what does it cost to implement the event predicate and uncertainty loading in a form suitable for fault-tolerant quantum computation?

This section provides two initial anchors for that feasibility discussion. First, we quantify how fixed-point precision affects collision-probability error in the encounter-plane computation, which informs how many bits are required for a reversible arithmetic implementation of the oracle predicate. Second, we report an initial resource-estimator sweep for a reversible oracle-arithmetic skeleton under representative fault-tolerant assumptions. These results are not intended as final resource claims; they are sanity checks that bound the magnitude of costs and clarify which design variables matter most.

7.1 Fixed-point precision requirements

Reversible quantum circuits are naturally expressed in fixed-point arithmetic. To avoid hand-waving about “enough precision,” we evaluate the error introduced by fixed-point quantization relative to a floating-point reference across representative benchmark scenarios. The resulting error-versus-bits curve provides a practical calibration: it identifies the precision range where quantization error becomes negligible relative to statistical uncertainty in the rare-event estimators, and it sets a defensible target for oracle arithmetic precision in subsequent resource estimates. Using the rule ‘choose the smallest fixed-point precision b such that quantization error $\leq 0.25 \times$ the estimator CI width,’ our sweep recommends **$b=12$** for baseline regimes and **$b=16$** for near-threshold stress cases.

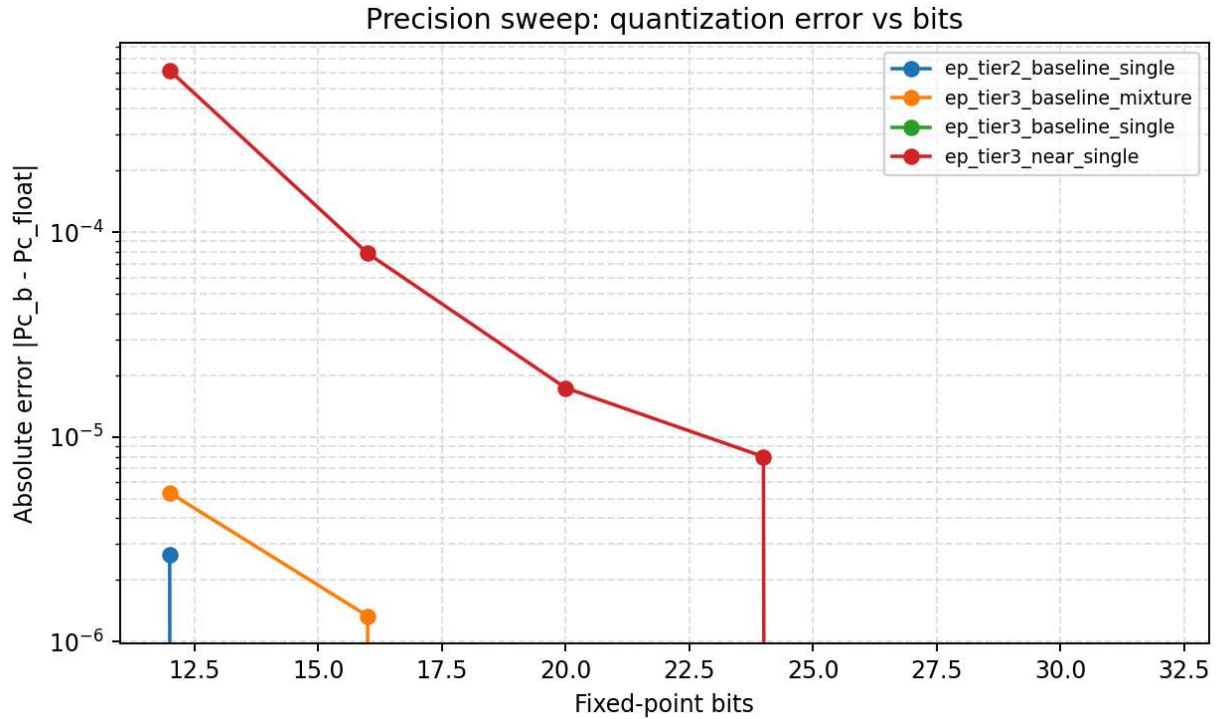


Figure 5. Absolute error between fixed-point and floating-point Pc computations versus fixed-point bit precision across representative scenarios. This provides an empirical handle on the precision required for reversible arithmetic implementations of the event predicate.

7.2 Initial fault-tolerant resource estimates (oracle skeleton)

We next report a preliminary QDK resource-estimator sweep for a reversible oracle-arithmetic skeleton. The sweep is intentionally simple: it does not claim a complete end-to-end implementation of a conjunction pipeline. Instead, it estimates the physical-qubit requirements associated with implementing the core arithmetic and predicate structure at varying precision levels, under two illustrative hardware models. The purpose is to make the feasibility conversation concrete and to expose which parameters dominate physical-qubit count as precision and error budgets tighten.

Our sweep reports physical-qubit requirements for a reversible oracle-arithmetic skeleton under two illustrative hardware models (a gate-based qubit model and a Majorana/topological-style model), varying fixed-point bit precision (16/24/32) and an estimator error-budget parameter ($10^{-2}/10^{-3}/10^{-4}$). These results should be interpreted as order-of-magnitude indicators, suitable for comparing scenarios and guiding Phase II engineering priorities (oracle simplification, state-preparation structure, and end-to-end resource accounting). In Figure 6, `qubit_gate_ns_e3` denotes the gate-based qubit model and `qubit_maj_ns_e4` denotes the Majorana/topological-style model (internal labels preserved from the sweep script)."

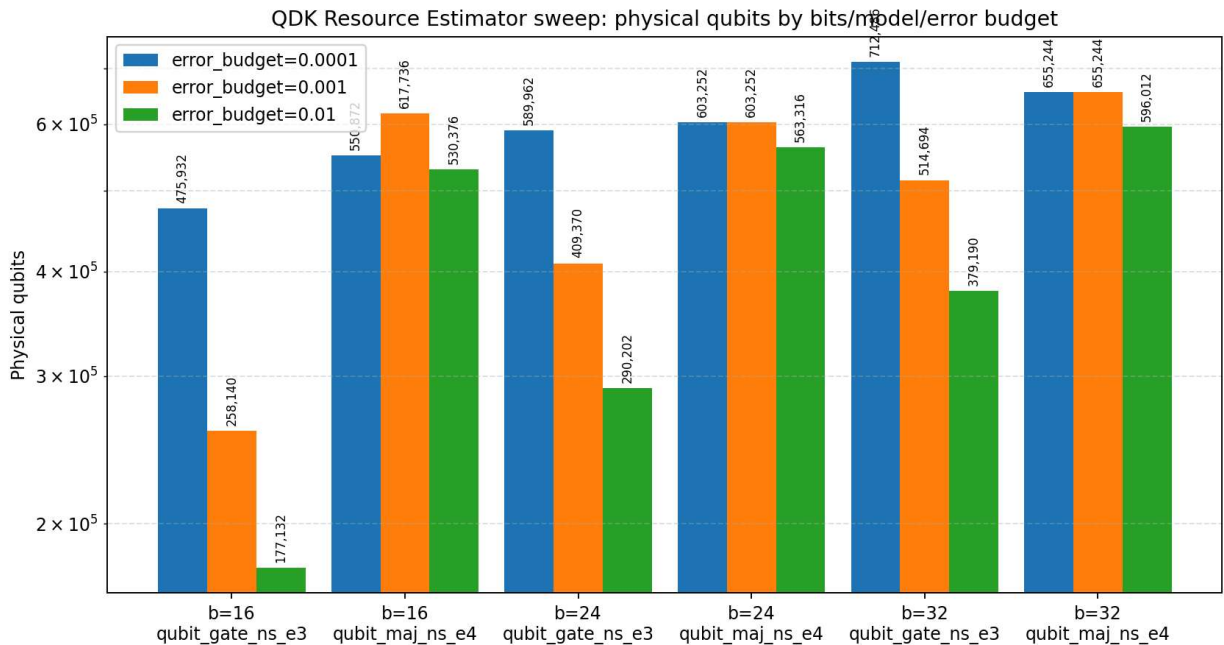


Figure 6. QDK resource-estimator sweep of physical-qubit requirements for a reversible oracle-arithmetic skeleton under two illustrative hardware models (gate-based and Majorana/topological-style). The sweep varies fixed-point bit precision (16/24/32) and an estimator error-budget parameter ($10^{-2}/10^{-3}/10^{-4}$), providing a concrete feasibility bound for subsequent Phase II refinement.

7.3 What this implies for the Phase II plan

Taken together, the precision calibration and the resource sweep motivate a straightforward Phase II engineering plan. First, treat oracle and state-preparation costs as first-class objects and explicitly account for them in any end-to-end advantage claim. Second, focus optimization effort where it moves feasibility the most: reducing arithmetic depth, tightening variable ranges, and structuring uncertainty models (including mixtures) so that state preparation and predicate evaluation remain tractable. Third, locate the “crossover regime” empirically by combining classical baseline curves (CI width vs sample budget) with fault-tolerant resource estimates, rather than assuming a crossover exists at operational scales today.

Limitations and scope

This report is intentionally scoped to a simplified, encounter-plane probability-estimation setting. We do not model full orbital propagation, maneuver dynamics, or end-to-end operational pipelines; rather, we isolate the statistical efficiency of rare-event estimators and the feasibility of a reversible event-predicate core under bounded precision. As a result, numerical values in this report should be interpreted as benchmark indicators rather than operational predictions for any specific mission. The fault-tolerant resource estimates are preliminary and reflect an oracle-arithmetic skeleton; they do not yet include a fully optimized state-preparation circuit or end-to-end compilation. These limitations are deliberate: they allow us to define a controlled benchmark, quantify time-to-confidence in threshold-sensitive regimes, and establish a transparent roadmap for Phase II work that expands resource accounting and validates robustness across broader scenario families.

8. Reproducibility

All figures and tables in this report are generated from the public repository:

<https://github.com/Xyloth/xpqa-rare-event-qae>

The following commands reproduce the benchmark runs and regenerate the core plots. Outputs are written to the same locations used by the report build (notably [paper/figures/](#), [results/plots/](#), and [results/tables/](#)).

```
python scripts/run_encounterplane_replicates.py && \  
python scripts/plot_encounterplane_paper_figures.py
```

```
python scripts/run_splitting_suite.py && \  
python scripts/plot_ciwidth_vs_compute.py
```

```
python scripts/generate_resource_estimate.py && \  
python scripts/run_qdk_resource_estimate.py --bits 16,24,32  
--error_budgets 1e-2,1e-3,1e-4
```

```
python scripts/run_gating_simulation.py && \  
python scripts/plot_gating_results.py && \  
python scripts/run_precision_sweep.py && \  
python scripts/plot_precision_sweep.py
```

What each line produces:

- Line 1: encounter-plane paper figures in [paper/figures/](#)
- Line 2: compute-normalized benchmark plot in [results/plots/ciwidth_vs_compute.png](#)
- Line 3: viability artifacts in [results/](#) and QDK sweep outputs in [results/tables/](#) and [results/plots/](#)
- Line 4: gating and precision sweep outputs in [results/plots/](#) (plus summary markdown in [results/](#))

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